

Research Report: YOLO-Based Visual Recognition Model and STM32 Deployment Preparation

1. Project Overview

Objective: To train a lightweight computer vision model capable of identifying "healthy" and "unhealthy" individuals within a flock of chickens, and to complete the preliminary preparation for deployment on the STM32N6570-DK.

Tools Used: Python, Anaconda, YOLOv8 (Ultralytics), PyTorch, ONNX, and ST Edge AI.

Target Hardware: STM32N6570-DK Development Board [1].

2. What is YOLO?

YOLO (You Only Look Once): YOLO is a single-stage object detection algorithm. Compared with traditional two-step detection methods that first identify an object and then determine its category, YOLO performs object localization and classification in a single process, enabling faster detection.

Why YOLOv8n: The suffix "n" stands for Nano, which is the smallest model in the YOLOv8 family. Since the final deployment environment is a microcontroller, it is necessary to select a lightweight model with a small number of parameters, low computational requirements, and fast inference speed.

3. Implementation Process

I) Environment Setup

In a pure CPU environment using an Intel i5 processor, an independent Python virtual environment (yolo_env) was established using Anaconda to prevent dependency conflicts.

During the configuration of Ultralytics and PyTorch, errors were encountered due to a small number of images in .heic/.heif format. The configuration was modified to skip the abnormal file formats, ensuring that the training process could proceed smoothly.

II) Model Training

Data Preparation: More than 7,600 poultry images were processed. Before training, clear data classification criteria were established. Chickens were classified as "unhealthy" based on abnormal physical characteristics such as lying down, messy or dirty feathers, exposed skin, and twisted postures. Path mismatch errors between the cloud-based dataset paths and the local hard-drive paths in dataset.yaml were also identified and fixed.

Model Training: The model was trained using YOLOv8n pre-trained weights for 3 epochs on a CPU. The model achieved an mAP@50 of 81.9%.

III) Performance Validation and Model Conversion

The model was tested using new images. It successfully detected the chickens, drew bounding boxes, and output "health" and "unhealth" labels together with confidence scores. The confidence scores were within the range of 0 to 1.

To adapt the model to STMicroelectronics' official toolchain, the PyTorch .pt model was converted into the ONNX (.onnx) format.

4. Summary

This project completed the main software-side preparation workflow for deploying an AI algorithm on an edge device.

Although the physical flashing process was paused due to the lack of the STM32N657 hardware, the .onnx model was successfully processed using the ST Edge AI Core toolchain, and the corresponding C-language neural network code (stai_network.c) was generated.

In addition, zero-error cross-compilation was successfully achieved in STM32CubeIDE, demonstrating that the software logic and firmware integration have been prepared for deployment.

5. Additional Information

Planned Deployment Process:

1. Use the ST Edge AI tool to process and optimize the current .onnx model and convert it into C-language code suitable for deployment on the NPU.
2. Use STM32CubeIDE to develop the firmware so that the STM32N6570-DK development board can automatically complete the entire process of capturing an image, sending it to the AI model for recognition, and outputting the result.
3. Flash the compiled program onto the microcontroller and test the actual operating speed and memory consumption under real-world conditions to verify that the software and hardware can work together properly.

[1] About the STM32N657 Development Board

The STM32N657 is a new generation of microcontroller featuring an integrated NPU. Its main processor is an ARM Cortex-M55, which operates at up to 800 MHz and supports Helium vector acceleration.

Compared with a conventional multi-core PC, it does not have the same level of general-purpose computing performance. However, it is specifically designed for edge computing and can efficiently process specific AI tasks with very low power consumption.

Its key hardware features include:

Independent Computing Power: The integrated NPU provides up to 600 GOPS of theoretical computing performance and is dedicated to neural network acceleration, significantly improving inference performance compared with conventional microcontrollers.

Ultra-Low Latency: The device integrates 34 MB of high-bandwidth SRAM and a CSI-2 camera interface supporting up to 5-megapixel resolution at 30 fps. This allows image data to be efficiently transferred for AI processing, enabling very low-latency visual feedback suitable for edge AI applications.

Hardware Offloading: The integrated ISP handles image processing tasks such as image noise reduction, further reducing the workload on the CPU.

The platform also features a NeoChrom 2.5D GPU and a 5-inch touchscreen display with an 800×480 resolution, providing responsive user interface capabilities.

Overall, the STM32N657 is well suited to this project because it combines low-power MCU operation with dedicated AI acceleration for edge deployment.

6. References

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Author's Declaration: The conceptual framework and dataset were provided by the supervising professor. The software engineering, model training, debugging, and ONNX conversion were carried out by the author and owner of this GitHub repository, with reference to relevant technical documentation and development resources.

基于 YOLO 的视觉识别模型与 STM32 的部署准备报告

1、项目概述

- 目标: 训练一个轻量级的计算机视觉模型, 能够在鸡群中识别出“健康 (health)”与“不健康 (unhealth)”的个体, 并完成在 STM32N6570-DK 上部署的前期准备。
- 用到的工具: Python, Anaconda, YOLOv8 (Ultralytics), PyTorch, ONNX, ST Edge AI。
- 参考硬件: STM32N6570-DK 开发板 [1]。

2、YOLO 是什么?

- YOLO (You only look once): 一种单阶段目标检测算法。对比传统算法“先识别目标, 再判断类别”的两步识别方法, YOLO 将图片划分网格, 一次性输出所有的物体位置和类别, 速度更快。
- 为什么选 YOLOv8n: 后缀的 n 代表 Nano (最小号模型) 这是因为我们最终的运行环境是单片机, 所以必须选择参数量最少, 计算量最小, 推理速度最快的轻量级模型。

3、具体流程

I) 环境搭建

- 在纯 CPU 环境下 (Intel i5), 使用 Anaconda 搭建了独立的 Python 虚拟环境 (yolo_env), 防止依赖冲突。
- 配置 Ultralytics 和 PyTorch 时, 遇到少量 .heic/.heif 格式的图片导致安装报错。通过修改配置跳过异常文件, 确保了训练流程顺利推进。

II) 模型训练

- 数据准备: 处理 7600+ 张禽类图片, 图片在训练前明确了数据分类标准 (如基于禽类躺卧、毛发凌乱肮脏、皮肤裸露、姿态扭曲等异常特征来定义“不健康”), 修复了 dataset.yaml 中云端路径与本地硬盘不匹配的报错。
- 模型训练: 基于 YOLOv8n 预训练权重, 用 CPU 跑了 3 个 Epoch 的训练。模型 mAP@50 准确率达到 81.9%。

III) 效果验证与模型转换

- 用新图片测试, 模型能成功识别并画出边框, 输出 health/unhealth 标签和置信度在0和1之间。
- 为适配 ST 官方工具, 把 PyTorch 的 .pt 模型转换成了 ONNX (.onnx) 格式。

4、总结

本项目走通了完整的 AI 算法落地的前置流程, 尽管由于缺少 STM32N657 硬件而暂停了物理烧录, 但我成功利用 ST Edge AI Core 工具链压缩了 .onnx 模型并生成了相应的 C 语言神经网络代码 (stai_network.c)。此外, 我在 STM32CubeIDE 中实现了零错误交叉编译, 证明软件逻辑和固件集成已完全准备就绪, 可以部署。

5、补充

实际部署流程:

- 利用 ST Edge AI 工具, 将目前的 .onnx 模型进行压缩, 转化为适配 NPU 的 C 语言代码。
- 用 STM32CubeIDE 写代码, 让 STM32N6570-DK 开发板能自动完成“拍照、传给 AI 识别、输出结果”这一整套动作。
- 把编译好的程序烧录到单片机里, 测一下真实环境下的运行速度和内存占用情况, 确认软件和硬件能完美配合。

[1]: 关于 STM32N657 开发板

这是一块内置 NPU 的新一代微控制器。主控 CPU 为 ARM Cortex-M55 (单核 800 MHz, 支持 Helium 矢量运算加速)。相比传统多核 PC, 它虽在通用算力上不具优势, 但专为边缘端设计, 能以极低功耗高效处理特定 AI 任务。其核心硬件特性包括:

- 独立算力: 内置 NPU 提供 600 GOPS 算力, 专注神经网络加速, 推理速度较普通单片机呈数量级提升。
- 超低延迟: 片内集成 34 MB 极高带宽 SRAM, 配合 CSI-2 摄像头接口 (支持 500W 像素/30fps), 图像数据可直通 NPU, 实现媲美甚至优于 PC 视觉反馈的极低延迟。
- 硬件减负: 内置 ISP 处理图像降噪, 进一步释放 CPU 算力; 搭载 NeoChrom 2.5D GPU 及 5 英寸触控屏 (800*480), 提供低延迟的 UI 响应。综上, 该硬件通过精简通用算力换取 AI 专精效能, 非常契合本项目的端侧部署需求。

6、参考资料

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